Q10) Pre-activation Z :-

$Z_{1,1} = (1\times1)+(2\times0)+(0\times-1)+(1\times1) = 2$

$Z_{1,2} = (2\times1)+(0\times0)+(1\times-1)+(1\times1) = 2$

$Z_{2,1} = (0\times1)+(1\times0)+(1\times-1)+(0\times1) = -1$

$Z_{2,2} = (1\times1)+(1\times0)+(0\times-1)+(2\times1) = 3$

$$Z = \begin{bmatrix} 2 & 2 \\ -1 & 3 \end{bmatrix}$$

ReLU Activation (Feature Map A) :-

$a_{1,1} = \max(0,2) = 2$

$a_{1,2} = \max(0,2) = 2$

$a_{2,1} = \max(0,-1) = 0$

$a_{2,2} = \max(0,3) = 3$

$$A = \begin{bmatrix} 2 & 2 \\ 0 & 3 \end{bmatrix}$$

Fully connected Layer (Y):

$A_f = [2, 2, 0, 3]^T$

$V = [1, -1, 2, 0, 5]$ bias $c = 1$

$Y = (V \cdot A_f) + C$

$Y = (1\times2)+(-1\times2)+(2\times0)+(0.5\times3)+1$

$Y = 2 - 2 + 0 + 1.5 + 1 = 2.5$

Loss (L) :- $T = 3$

$L = \frac{1}{2}(Y-T)^2 = \frac{1}{2}(2.5-3)^2$

$= \frac{1}{2}(-0.5)^2 = \frac{1}{2}(0.25) = 0.125$

Q1 b) Backward pass :-

$$\frac{\partial L}{\partial y} = (y - T) = 2.5 - 3 = -0.5$$

$$y = V \cdot A_f + C \quad \therefore \frac{\partial y}{\partial V} = A_f^T$$

$$\frac{\partial L}{\partial V} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial V} = -0.5 \cdot [2, 2, 0, 3]$$

$$\frac{\partial L}{\partial V} = [-1, -1, 0, -1.5]$$

$$\frac{\partial L}{\partial A_f} = \frac{\partial L}{\partial y} \cdot V = -0.5 \cdot [1, -1, 2, 0.5]$$

$$= [-0.5, 0.5, -1, -0.25]$$

$$\therefore \frac{\partial L}{\partial z_{1,1}} = -0.5$$

$$\frac{\partial L}{\partial z_{1,2}} = 0.5$$

$$\frac{\partial L}{\partial z_{2,1}} = 0$$

$$\frac{\partial L}{\partial z_{2,2}} = -0.25$$

$$\frac{\partial L}{\partial W} = X * \frac{\partial L}{\partial z}$$

$$\frac{\partial L}{\partial w_{1,1}} = 0.25, \quad \frac{\partial L}{\partial w_{1,2}} = -1.25, \quad \frac{\partial L}{\partial w_{2,1}} = 0.5, \quad \frac{\partial L}{\partial w_{2,2}} = -0.5$$

$$\frac{\partial L}{\partial W} = \begin{bmatrix} 0.25 & -1.25 \\ 0.5 & -0.5 \end{bmatrix}$$

Q1c) $W_{1,2}$ has largest absolute gradient value 1.25. GD updates weight as $W_{new} = W_{old} + \eta \frac{\partial L}{\partial W}$ and gradient is negative, this weight increased the most.

Q3a) Max Pooling:-

New Y : max (A) = max (2, 2, 0, 3) = 3

Gradient Flow: only the winning unit $a_{2,2}$ passes through a gradient of 1, others are 0.

Zero Gradient: $x_{1,1}$, $x_{1,2}$, $x_{1,3}$, $x_{2,1}$ have 0 gradient as they donot contribute to $a_{2,2}$.

Q3b) Normalization:-

$$\mu = \frac{2+2+0+3}{4} = 1.75$$

$$\sigma^2 = \frac{(2-1.75)^2 + (2-1.75)^2 + (0-1.75)^2 + (3-1.75)^2}{4}$$

$$= 1.1875$$

$$A_{norm} = \frac{A - 1.75}{\sqrt{1.1875}} \approx \begin{bmatrix} 0.229 & 0.229 \\ -1.606 & 1.147 \end{bmatrix}$$

Q 3c) Translation Variance :-

Shifting X will move the high activation within A. Global max pooling only selects the max value regardless of its coordinates. Y remains 3, leaving the loss unchanged.